

Zijian (Jack) Luo

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Education

Monash University <i>PhD in Computer Science</i>	Oct 2026 – Present <i>Australia</i>
The University of Sydney <i>MPhil in Computer Science</i>	Mar 2024 – Mar 2026 <i>Sydney, Australia</i>
Hebei University <i>B.Eng. in Software Engineering</i>	Sep 2018 – Jun 2022 <i>Hebei, China</i>

Work Experience

 Microsoft Research Asia (MSRA), Microsoft <i>Research Intern</i>	Mar 2026 – Present
- Enhanced Copilot's cross-platform code migration capability by integrating a knowledge base to model platform-specific APIs, dependencies, and migration mappings. - Built a dependency-upgrade benchmark for long-horizon software engineering tasks to evaluate MAI Coder's end-to-end planning and execution across multi-step workflows; the corresponding paper is under review. - Collaborated with GitHub to propose leveraging coding-agent development trajectories as process-aware context for Copilot Code Review (CCR) to better understand code changes; the work is under review.	
 Oracle Labs, Oracle <i>Research Assistant</i>	May 2025 – Oct 2025
- Conducted research on software supply-chain reliability by integrating LLMs with the Macaron framework, reproducing build artifacts for over 120 open-source repositories from Libraries.io with an 80% success rate. - Designed LLM-assisted workflows for defect reproduction and static-analysis validation.	

Publications

- Update from Hell: Can Coding Agents Survive Hidden Breakage in Dependency Upgrades?**
Zijian Luo, Runzhi He, Pengfei Gao, Yu Kang, Zeqi Lin, Minghua Ma, Qingwei Lin, Saravan Rajmohan, Yongqiang Tian.
arXiv preprint arXiv:2608.30300, 2026.
- Automatic Data Repair without Format Specifications.**
Zijian Luo, Lukas Kirschner, Ezekiel Soremekun, Rahul Gopinath.
IEEE International Symposium on Software Reliability Engineering (ISSRE) 2025. CCF B CORE A
- Assessing Reliability of Statistical Maximum Coverage Estimators in Fuzzing.**
Danushka Liyanage*, Nelum Attanayake*, **Zijian Luo***, Rahul Gopinath.
IEEE International Conference on Software Maintenance and Evolution (ICSME) 2025. CCF B CORE A
- Maximal Format-Free Record Repair.**
Zijian Luo, Xi Wu, Hong Jin Kang, Alan Fekete, Rahul Gopinath.
IEEE/ACM International Conference on Automated Software Engineering (ASE) 2026. CCF A CORE A*

Selected Projects

- The Deeplang Programming Language (82 Stars)**
A memory-safe, multi-paradigm programming language for embedded systems. I was responsible for implementing the backend virtual machine, including interpreting and executing WebAssembly (WASM) code generated by the frontend.

Technical Skills

Programming Languages	C/C++, Python, Rust, WebAssembly
Analysis & Testing	Fuzzing (AFL), Program Repair, CodeQL
AI	LLMs, SFT, RL